

FAN, Sixing

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Education

CUHK, BSc in Mathematics and Information Engineering Sep 2023 – Present

- S.H.Ho College Master's List Award 2025
- Faculty of Engineering Dean's List 2024/2025
- S.H.Ho College Matriculation Scholarship for Academic Excellence 2023/2024

Research Experience

Safety-Constrained Dual-UAV Visual Tracking Simulation, CUHK IE Department Jun 2026 – Present

- Built ROS/PX4/Gazebo dual-UAV vision-tracking simulation architecture
- Integrated YOLO, Kalman filtering, timestamp alignment, and velocity control
- Implemented centralized safety supervision with collision constraints

Robot Indoor Navigation, CUHK IE Department Jun 2025 – Feb 2026

- Formulated Static Scene Complexity Score (SSCS) to evaluate the indoor scene complexity
- Developed a SSCS optimizer for generating indoor scenes of various scene complexity using Python
- Conducted navigation experiments using scenes generated by SSCS optimizer in a simulated environment

Projects

CUHK MailRoute: Opportunity Discovery Platform [\[Demo\]](#) Jul – Aug 2026

- Deployed a bilingual opportunity recommendation system ranking CUHK mass emails
- Developed a React 19 TypeScript PWA using Tailwind CSS
- Automated digest collection, parsing, validation, and structured repository updates

Kaggle Orbit Wars: Strategic Game-Agent Development May – Jun 2026

- Built a Top 10% Python agent by heuristics for this 2/4-player strategy game
- Developed a local arena and replay-analysis workflow to diagnose strategic failures
- Prototyped imitation learning and PPO reinforcement learning pipelines

Gradient-Based Adversarial Attacks: FGM, FGSM and PGD Nov 2025

- Investigated three gradient-based adversarial attack methods through mathematical formulation
- Conducted empirical evaluation by Robust Bench framework on ImageNet-1k validation set using ResNet-18
- Demonstrated that PGD exhibits the strongest attack capability

Internship Experience

IT Support, Tech Vision System Limited, Hong Kong Jun – Aug 2024

- Completed a project of Huawei IdeaPresence according to customer requirements
- Configured products such as Huawei IdeaHub S2, ES2, ES2 Plus
- Edited and updated the company website and operated social media of the company
- Provided technical support and troubleshooting for IT infrastructure.

Skills

Languages: Mandarin (Native), English (Fluent), Cantonese (Fluent)

Programming: Python, Java, C, MATLAB, TypeScript and LaTeX

Frameworks: PyTorch, NumPy, Matplotlib, pandas, bpy, scikit-learn and xgboost

Tools: VS Code, Cursor, GitHub, Hugging Face, Blender, Linux System, Wolfram Alpha and Wireshark/tcpdump

Mathematics: Calculus, Linear Algebra, Abstract Algebra, Complex Analysis, Discrete Mathematics and Probability